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Pollution in the Pilcomayo: Mining and Indigenous Communities

Case Description

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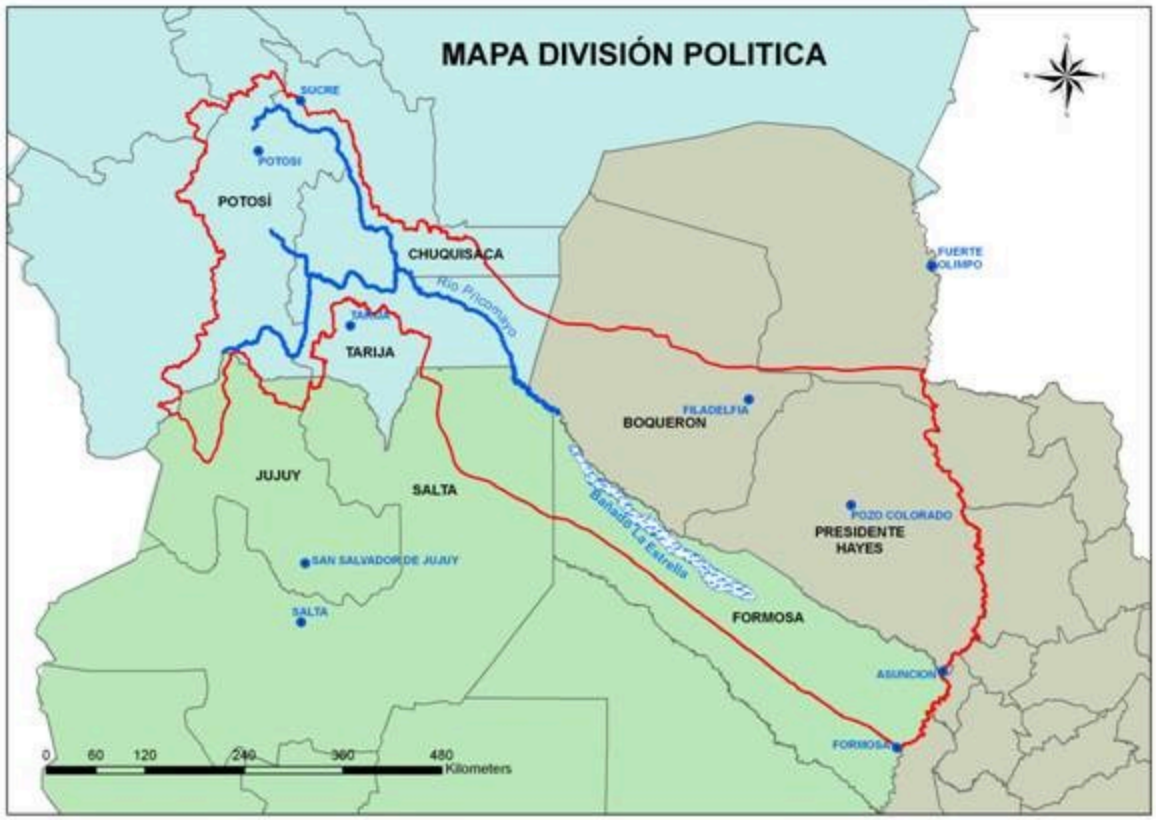
Geolocation:	22° 2' 17.24" S, 62° 41' 0.94" W
Total Population:	million
Climate Descriptors:	Humid mid-latitude (Köppen C-type)
Predominent Land Use Descriptors:	agricultural- cropland and pasture, conservation lands, mining operations, rangeland, religious/cultural sites
Important Uses of Water:	Agriculture or Irrigation,Fisheries - wild,Mining/Extraction support
Water Features:	La Plata Basin; parana;
Riparians:	Bolivia; Paraguay; Argentina;
Agreements:	Agreement constituting the trilateral commission for the development of the Pilcomayo river basin

Summary

The Pilcomayo River is experiencing considerable pollution from mining activities in Bolivia, affecting the downstream riparians Argentina and Paraguay. The pollution is problematic in particular for indigenous communities who live from the river, and studies have found symptoms of heavy metal exposure among them. Even though the three countries have an exemplary framework for joint basin management, the pollution issue has so far not been addressed. The key reason is that indigenous communities are not represented by their national governments, shifting the conflict to the domestic level. Broader stakeholder participation could be an effective measure to bring the conflict to the table and developing the tourism industry in Bolivia could help to resolve it.

Natural, Historic, Economic, Regional, and Political Framework

Background



Geography of the Basin

The Pilcomayo river basin is approximately 1000 km long, connecting Bolivia, Argentina, and Paraguay, covering an area of 290'000 km2. The headwaters of the Pilcomayo are in Bolivia, and for the first 500 km the Pilcomayo is a mountain river that drops from an elevation of over 4000m to an elevation of 270m at Villa Montes, close to the point where it leaves Bolivia and forms the border between Paraguay and Argentina. From there, the river flows through the Gran Chaco Plain and has an extremely low gradient.

In principle, the Pilcomayo joins the Rio Paraguay at Asuncion, contributing to the Parana and eventually the La Plata river. However, the waters of the Pilcomayo do not actually cross the Gran Chaco. Instead, the river stops roughly halfway between Villa Montes and Asuncion, forming floodplains and wetlands where the water evaporates or seeps into the groundwater. Some water that does eventually flow into the Rio Paraguay comes from rainfall in the Gran Chaco plain itself, it has no hydrological connection to the Pilcomayo

The reason for this unusual situation is that the Pilcomayo carries an extremely high sediment load, reaching at times 60g of sediment per liter (Martín-Vide et al. 2014). In combination with the low gradient in the Gran Chaco, the Pilcomayo is the only river in the world that regularly blocks itself with debris and sediment during the flood season. As a result, the water spills over to find new flow paths, changing the course of the river and creating temporary wetlands in the process, the largest of which is referred to as the Banado la Estrella. This endpoint of the main river channel has been moving upstream over the course of the last century.

Partly as a result of this unusual geomorphology, the Gran Chaco is a biodiversity hot-spot. The seasonal wetlands support a wide range of mammals, fish, reptiles, and birds, many of them rare or endangered, such as the giant armadillo (Alcorn et al. 2010; Braunstein et al. 2003).

Human Population of the basin

The basin is home to an estimated 1.5 million people. A significant share of the population in the basin is indigenous, prominent among them are the Wichi and the Toba, but there are at least 20 distinct indigenous communities. In Bolivia, they make up 37% of the basin population, in Paraguay 32%, and in Argentina 7% (Laboranti 2011).

The people living in the Pilcomayo river basin are relatively poor. There are no cities with more than 100’000 inhabitants except Potosi in Bolivia at the very beginning of the river and Asuncion in Paraguay at the very end of it. According to estimates, 60% of the population live below the poverty lines of their respective countries, and roughly 20% live under conditions of extreme poverty (Laboranti 2011).

Economy of the basin

The main economic activities in the basin are mining and agriculture. Potosi, Bolivia, is a significant silver mining area, founded over 500 years ago by Spanish imperialists and still productive today. There is some irrigated agriculture and commercial production of soy and livestock. Most of the agriculture, however, is subsistence farming and herding. Fishing, focused on the species Sabolo, is an important economic activity in the indigenous communities. The overall significance of the Pilcomayo river basin for national economies in terms of GDP share is 12.2% in Bolivia, 4.7% in Paraguay, and 0.5% in Argentina (Laboranti 2011).

Conflict

The Pilcomayo basin is grappling with a number of issues that have conflict potential. First, mining operations in Bolivia pollute the Pilcomayo river, which affects the environment and humans downstream (Stassen et al. 2012). Second, the high sediment load causes the river to frequently change course while flowing through the Chaco Plain. As a result, it is unpredictable how much water Paraguay and Argentina will receive every year (Martín-Vide et al. 2014). Third, there is a severe ecological crisis in the lower Pilcomayo (Sidder 2016), the causes of which are unclear and give rise to

controversies. And fourth, the region is said to have weak institutions who do not reliably fulfill their responsibilities in terms of water management and law enforcement (Vera 2016; Strosnider et al. 2011).

All of these issues are related and contribute to the complex situation of the Pilcomayo river basin. This case study focuses on the issue of pollution from mining activities, analyzes in which ways it leads to conflict and how these conflicts could be solved.

Pollution Issue

Mining is an important activity in the Bolivian Andes, where the Pilcomayo originates. Over 500 years ago, the Spanish discovered the "Cerro Rico", a mountain with extraordinary mineral wealth, containing the world's largest known silver deposit. The Spanish founded the city of Potosi and the silver that was produced there contributed significantly to the wealth of the Spanish Empire. Ever since, people in the region have been mining silver, tin and other metals, yet even today the mountain's mineral reserves have not been exhausted.

The mining activities have released large amounts of toxic substances with effects on downstream water users. There is evidence of a consistent flow of acid mine drainage that originates both from active and abandoned mine sites. This acid mine drainage contains eco-toxic metals such as Aluminum, Arsenic, Cadmium, Copper, Iron, Magnesium, Lead, and Zinc, some of them in concentrations that are orders of magnitude above discharge limits set by the Bolivian government (Strosnider et al. 2011). In addition to the constant flow of pollutants, there were at least two significant spills of mine tailings, sending pulses of pollutants down the river, once in 1996 at the Porco mine (Edwards n.d.), and in 2014 at the Santiago Apostol mine (Donoso 2014).

The full extent and the exact consequences of this pollution from mining are uncertain, because there is no systematic sampling and monitoring system in place. Nevertheless, it is very likely that downstream water users in Bolivia, and also further downstream in Argentina and Paraguay are exposed. There is evidence that indigenous communities, who eat a lot of fish from the Pilcomayo river, have elevated concentrations of lead in their hair (Stassen et al. 2012; Safi 2004). Also, families in these communities are smaller and children learn how to walk later, when compared to a control group that is not exposed to mining pollution upstream.

In sum, mining in Potosi, Bolivia causes pollution in the Pilcomayo river, affecting people as well the environment downstream. This sets up a conflict between Bolivia and the two downstream countries Paraguay and Argentina.

Key Stakeholders

Mining cooperatives

Most of the mining in the basin is undertaken by small mining companies and cooperatives close to the city of Potosi, Bolivia, a city with roughly 250'000 inhabitants. According to official estimates, there are 450 small scale mining operations, 80% of which do not have a valid environmental license (Donoso 2014). International players such as the World Bank and the British mining company Rio

Tinto Zinc used to have stakes in mining operations in Potosi in the 1990ies, but seem to have exited the business since. Many, if not most, of the people who work in the mines have indigenous origins. The mining industry has a long history in Potosi, and is the only industry in the area that provides jobs.

Indigenous communities

The Pilcomayo river is home to many indigenous communities. Given that they make up roughly a fourth of the basin population (Laboranti 2011), a careful estimate for the size of the indigenous population would be 200'000 to 400'000 people. The largest communities are the Toba, the Pilagá, and the Wichí (Braunstein et al. 2003), but there are over twenty different indigenous communities. Since many of them rely on the river for drinking water and for fishing, they are most directly affected by pollution in the river. Many of them live in Bolivia, where the effects of pollution are likely to be most severe, but more than half of them live in Argentina and Paraguay in the Gran Chaco.

The tri-national commission

Bolivia, Paraguay, and Argentina have formed a trilateral commission that is charged with surveying and managing the basin, as well as overseeing its sustainable development.

International donors

Development in the Pilcomayo river basin have been influenced strongly by international donors such as the UNDP, the Inter-American Development Bank (IDB) and the European Union. Over the decades, the have financed development efforts and influenced river basin management practices.

History of the conflict and conflict resolution

The conflict over pollution is remarkably silent. The riparian countries have developed an elaborate framework for joint management and conflict resolution, yet none of the countries is using this framework to address the pollution issue. This section describes how the collaborative framework evolved and how it deals with pollution. The following section evaluates the framework and investigates why the pollution problem has not been addressed so far.

Argentina, Bolivia, and Paraguay have a joint commission with the task of studying and coordinating development opportunities in the Pilcomayo Basin. This commission was first formed as a result of a 1941 agreement on the utilization of the waters of the river Pilcomayo (Argentina et al. 1941). Similar to the La Plata treaty of 1967 (which included the additional parties of Brazil and Uruguay), the agreement and the work of the commission was focused on exploring the river's potential for dams, hydropower and irrigated agriculture.

In 1975, the UNDP and the IDP initiated and funded a comprehensive study of the Pilcomayo river basin (OAS 1979). While the topic of pollution was within in the scope of the study, it produced no assessment or recommendation on the matter. Instead, it focused on identifying potential hydropower and agricultural development projects. The study report also makes no mention of indigenous populations or biodiversity.

In 1994, the three riparian countries signed another treaty (Bolivia et al. 1994), modeled on the La Plata basin treaty, which created a new trilateral commission that would collaborate even closer. Specified as tasks of the commission are among others (1) proposing rules for the discharge of pollutants, and (2) monitoring water quality systematically and informing the parties of infringements (see Article 4, paragraph j and k). In addition, the treaty lays a basis for far-reaching technical collaboration and data-sharing. It specifies that decisions are to be taken unanimously and that conflicts are escalated to the heads of state if necessary.

Between 2000 and 2008, the commission prepared a comprehensive basin masterplan and produced a great deal of studies and reports which are for the most part publically available on the webpage www.pilcomayo.net. This effort was funded by the European Union with EUR 12.6 million, matched with EUR 3.9 million from the three basin countries (Laboranti 2011). The overall objective of the masterplan (Comision Trinational 2005), finalized in 2005, was to improve the quality of life of the basin residents. Among its many objectives was the monitoring of water quality, even with a specific view to the pollution generated by mining activities (see section 4.3.V).

However, the resulting activities in the years after the drafting of the masterplan did not amount to an adequate monitoring of the mining pollution problem. The commission's library provides one report that measures metal concentrations in mine effluent around Potosi (Peñaloza & Reinhard 2000), and a master's thesis that studied related public health outcomes(Safi 2004), both of which predate the masterplan. The annual water quality bulletins that are published on the commission's webpage are limited to generic water quality parameters that do not speak to the potentially dangerous metal ion concentrations. Some of the annual water quality reports state that samples are currently analyzed for metal concentrations, but no such data is eventually reported. Several reports that might be relevant to the issue, such as the "Quality of the Aquatic Environment, 2009", are not publically available. In summary, many independent studies that have investigated pollution and eco-toxicity in the Pilcomayo indicate that it is a grave problem. Yet, the commission so far does not seem to collect data systematically, despite massive financial and technical support from international donors.

Also in the minutes of the tri-annual commission meetings, pollution issues have been conspicuously absent for the most part, even though this may be changing. The tailings spill caused in June 2014 by the Santiago Apóstol mine (Donoso 2014) was addressed at the following meeting of the commission on October 23, 2014. The Bolivian delegation committed to deliver a detailed report (which is not publically available), while the Argentinian delegation called for the establishment of emergency response protocols. While these official statements sound fairly benign, the case was taken up in Bolivian courts, and three years later the owner of the mining company was sentenced to seven years in prison (La Razon 2017). This is the first time that a mining operative in Potosi has been punished for causing pollution.

Outcomes and Evaluation

In many ways, the degree of collaboration between the three riparian countries is examplary: the three countries share data and communicate transparently, the commission has a political, a technical, and a participatory arm, meetings are held frequently, and there are provisions for conflict resolution and amending the agreement when circumstances change.

The outcomes, however, have been less impressive. As outlined in the previous section, the pollution issue has largely been ignored. But the commission also had very limited success with building infrastructure in the basin, which had been one of the core objectives of collaboration from the start. The Pilcomayo river flows undammed, produces no hydropower, and irrigates a relatively small growing area (Brogan 2012). The explanation is perhaps that the Pilcomayo with its enormous sediment load is very difficult to develop. On the upside, the riparian countries have never had strained relationships over water issues and they have gathered a great deal of knowledge about the hydrology of the river. It seems that the trilateral commission would have the institutional and technical capacity to address the pollution problem – and so the question is why it has been reluctant to do so.

It is somewhat risky to explain this without speaking directly with the people involved. However, based on the available literature, three reasons seem plausible. First, Bolivia has a motif to protect its mining industry, which supports a great deal of workers and is a matter of national priority. Second, indigenous populations are not well-represented by their national governments, protecting indigenous livelihoods does not seem to be a national priority in any of the three basin countries. Third, the consequences of heavy metal pollution are slow-acting and hard to see, whereas other problems such as malnourishment are immediate and obvious, so that even within affected communities, pollution is not at the top of the agenda.

Most interesting from a water diplomacy perspective is the second reason, namely that important stakeholders of the basin are not participating in the negotiation process. While the commission has created a mechanism to consult indigenous communities while working with the European Union (Laboranti 2011), some observers claim that these mechanisms are ineffective. Allegedly, the experts charged with developing the master plan listened to the observations of indigenous communities but eventually ignored them (Alcorn et al. 2010). In that way, knowledge that could have informed effective development measures in the basin was lost. For example, the production of the native algarrobo may be more appropriate in the basin than the production of soy (Alcorn et al. 2010).

At a more general level, the reason is that transnational water conflicts can shift to national arenas, when stakeholder participation fails. In comparison to other continents, water disputes in Latin America are much less confrontational (Biswas 2011). Yet, considering that the displacement and the impoverishment of indigenous communities is an important problem in many countries in Latin America, the truth may be that water disputes seem less confrontational, because the real disputes are not brought to the international arena. This shift of disputes from the international negotiation to domestic politics is a likely explanation why the pollution problem in the Pilcomayo case stays under the radar and does not lead to an international conflict.

Recommendations

Broadly, two steps will be necessary to resolve the pollution problem in the Pilcomayo River Basin. First, the downstream countries need to bring the conflict to the table by involving a greater number of stakeholders, predominantly indigenous communities, but also conservation NGOs and local politicians. Second, to then also resolve the conflict, a wider focus on value creation that goes beyond hydropower, mining, and irrigated agriculture holds promise.

The first recommendation is to involve more stakeholders in the process that is led by the commission. There are several reasons to believe that this will improve the outcomes of its negotiation and planning efforts. First, the basin concerns a relatively small part of each country, in terms of population, area, and economic importance. Large projects of national significance, which were initially hoped for, are unlikely to materialize at this point. Therefore, a greater weight for provincial governments in the commission could tie the commission's work closer to the needs of the people who live in the basin. Second, NGOs are becoming stronger in the area, building networks among indigenous communities and conducting independent monitoring campaigns (Alcorn et al. 2010). Their contribution of knowledge and data at low cost could enhance the work of the commission. Given that many of the more centralized and large-scale objectives of the commission have not materialized over the last four decades, this information might provide the commission with proposals for smaller interventions. If the pollution problem is not brought up by those stakeholders, the commission might consider awareness campaigns in order to prevent the greatest health hazards, such as lead intoxication for breastfed infants.

As for the second recommendation, the Pilcomayo Basin has been handed a golden opportunity to counterbalance a squeeze of the mining industry with an expansion into tourism. The UNESCO has recognized Potosi as a cultural heritage site, which offers an enormous opportunity for a city of 250'000 inhabitants to become a popular tourist destination. Already, a 500-year history of mining and the prospect to visit the tunnels of "Cerro Rico" attract numerous backpackers. However, the UNESCO has warned that it may retract the cultural heritage status, worrying that illegal mining might destabilize the mountain. This provides a strong incentive for Bolivia to get a grip on illegal mining operations, while also opening up alternative employment opportunities.

Paraguay and Argentina should support the development of tourism in Potosi, because it will change Bolivia's incentives structure regarding pollution and illegal mining. In addition, also the Gran Chaco is of touristic value, with its cultural and ecological diversity, and might be able to profit from the vicinity to Potosi. As a first step, and in connection to the first recommendation, representatives of the tourism industry should participate in the commission's work.

Outlook

For five centuries, mining activities have polluted the Pilcomayo river. For seven decades the three riparian countries have tried to develop the basin economically, but no major infrastructure has been built. The situation of indigenous communities is dire and the basin in general is poor. However, things may be going in a good direction now. The commission has begun to open up to a wider range of stakeholders, even if this process still needs to mature. The opportunity to develop tourism changes the calculus of Bolivia vis-à-vis its mining industry. And the recent conviction of a mining operative for pollution is a signal that there is some political will in Bolivia to crack down on mining pollution. It is a good moment for the three countries, as well as for the various stakeholder groups within them, to develop their basin in a new direction.

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Issues and Stakeholders

Pollution

NSPD: Water Quality
Stakeholder Types: Sovereign state/national/federal government, Industry/Corporate Interest, Community or organized citizens

There is significant pollution with heavy metals from mining operations in Bolivia, affecting water users downstream in Argentina and Paraguay, in particular indigenous communities.

The key stakeholders are mining cooperatives upstream, indigenous communities who live by and off the river, the tri-national commission through which Bolivia, Paraguay, and Argentina manage the basin, and international donors who aim to aid the basin's development.

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Key Questions

Power and Politics: ***To what extent can international actors and movements from civil society influence water management? How and when is this beneficial/detrimental and how can these effects be supported/mitigated?***

Several important donors have been active in the Pilcomayo Basin. Also, there are budding initiatives by indigenous groups to organize. The question is, to what extent these forces from above and below are effective in asserting tighter control over pollution.

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